

Student Performance Prediction Using Machine Learning

Formative Assessment 2 Report – Artificial Intelligence & Machine Learning Mini Project

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Abstract—Educational institutions frequently identify academically at-risk students only after final examination results are declared, leaving limited scope for timely intervention. This mini project implements and evaluates supervised machine learning models to predict whether a student will pass or fail in a secondary school mathematics course using the publicly available UCI Student Performance dataset. The work follows a complete pipeline including data understanding, preprocessing, feature engineering, model training, hyperparameter tuning, evaluation using multiple metrics, and result interpretation. Logistic Regression, Decision Tree, and Random Forest classifiers are implemented, and a tuned Random Forest model is developed using grid search. Experimental results show that data-driven models can achieve competitive performance while also providing interpretable insights into influential academic and behavioral factors. The study demonstrates how AI and ML techniques can be applied in a real-world educational context to support early warning systems and formative assessment.

Index Terms—Student performance prediction, machine learning, Random Forest, Logistic Regression, educational data mining, formative assessment.

I. INTRODUCTION

Formative assessment in modern education increasingly relies on data-driven insights to monitor learner progress and support timely remedial action. Traditional evaluation methods, such as periodic tests and manual teacher observation, are often reactive, time-consuming, and limited in their ability to capture complex patterns across multiple variables. As a result, educators may detect struggling students only after significant performance deterioration, which negatively impacts academic outcomes and increases dropout risk.

Recent advances in artificial intelligence (AI) and machine learning (ML) have enabled predictive models that can infer future academic performance from historical and behavioral data. By leveraging features such as socio-demographic information, study habits, previous grades, parental background, and co-curricular activity participation, ML models can estimate the probability of success or failure for individual students. These predictions can then inform targeted interventions, resource allocation, and personalized guidance.

This mini project, carried out as part of Formative Assessment 2 (FA2) for the AI & ML course in the academic year 2025–26 at Pimpri Chinchwad College of Engineering (PCCoE), focuses on predicting secondary school student performance using classical ML models. The project uses the well-known Student Performance dataset from the UCI Machine Learning Repository, restricted to the mathematics course. The target variable is formulated as a binary outcome representing pass or fail, and the work explores the effectiveness of Logistic Regression, Decision Tree, and Random Forest classifiers.

The key contributions of this report are:

- Design of an end-to-end ML pipeline for student performance prediction, including data preprocessing, feature encoding, and model selection.
- Comparative evaluation of multiple models using accuracy, precision, recall, F1-score, and ROC-AUC metrics.
- Hyperparameter tuning of the Random Forest classifier using grid search to improve predictive performance.
- Analysis of feature importance and correlation to identify key factors associated with student success and failure.

The remainder of this report is organized as follows. Section II presents a brief literature review. Section III describes the dataset. Section IV details the methodology and ML pipeline. Section V discusses experimental results. Section VI concludes the report and outlines future work.

II. LITERATURE REVIEW

Student performance prediction has been widely studied within the broader area of educational data mining. Early work by Cortez and Silva applied decision trees, random forests, neural networks, and support vector machines to the UCI Student Performance dataset to predict final grades in mathematics and Portuguese language courses [1]. Their results demonstrated that ensemble methods such as Random Forest and Gradient Boosting can achieve strong predictive accuracy by combining information from demographic, social, and academic attributes.

Subsequent studies have extended this line of research in several directions. Certain authors have focused on applying deep learning architectures, such as multilayer perceptrons and recurrent neural networks, to handle high-dimensional and temporal data in virtual learning environments. Other work emphasizes interpretability by using decision tree-based models and explainable AI techniques to identify important predictive features such as prior failures, absences, and parental education. There is also growing interest in integrating real-time learning analytics from learning management systems (LMS) and online assessments.

Despite progress, many existing studies report limitations related to small sample sizes, limited generalizability across institutions, and a lack of deployment-ready systems. In addition, some models are treated as black boxes, making it difficult for educators to understand the reasoning behind predictions. This mini project addresses these concerns at a small scale by:

- Using interpretable baseline models alongside ensemble methods;
- Reporting a comprehensive set of evaluation metrics;
- Highlighting feature importance and correlation analysis relevant for educators.

III. DATASET DESCRIPTION

A. Source

The project uses the *Student Performance* dataset from the UCI Machine Learning Repository, which contains data about Portuguese secondary school students in mathematics and Portuguese language courses [2]. The current work focuses on the mathematics course file `student-mat.csv`, which consists of 395 instances and 33 attributes.

B. Attributes

The dataset includes demographic, social, academic, and behavioral features. A non-exhaustive summary is provided below:

- **Demographic:** school (GP/MS), sex (F/M), age, address (urban/rural), famsize (family size), Pstatus (parental cohabitation).
- **Parental background:** Medu, Fedu (mother's and father's education), Mjob, Fjob (parental occupations).
- **Academic:** studytime, failures (past class failures), schoolsup (extra educational support), paid (extra paid classes), G1, G2, G3 (first, second, and final period grades).
- **Behavioral and social:** activities, internet, romantic, famrel, freetime, goout, Dalc (workday alcohol), Walc (weekend alcohol), health, absences.

C. Target Variable

For this project, the continuous final grade G3 (ranging from 0 to 20) is converted into a binary classification label:

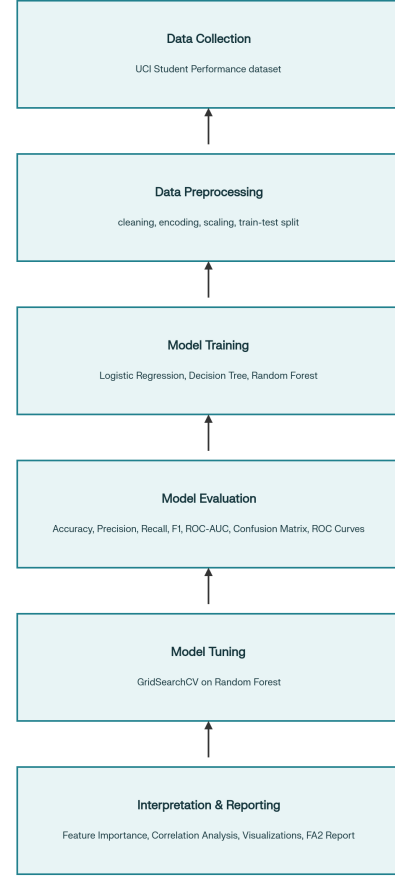


Fig. 1. End-to-end machine learning pipeline for student performance prediction, from data collection to deployment-ready model.

$$\text{target} = \begin{cases} 1, & \text{if } G3 \geq 10 \text{ (Pass)} \\ 0, & \text{if } G3 < 10 \text{ (Fail)} \end{cases}$$

After target creation, G3 and the intermediate grades G1 and G2 are removed from the input features to avoid data leakage and to place stronger emphasis on demographic and behavioral indicators.

The resulting dataset thus contains 395 samples with 30 input features and one binary target variable. The pass rate is approximately two-thirds of the cohort, indicating moderate class imbalance.

IV. METHODOLOGY

A. Overall Pipeline

The project follows an end-to-end ML workflow as depicted in Fig. 1. Major stages include data loading, preprocessing, model training, evaluation, and interpretation.

B. Data Preprocessing

The preprocessing stage consists of the following steps:

- 1) **Missing value analysis:** The dataset is checked for missing values and is found to be complete, therefore no imputation is required.
- 2) **Target encoding:** The binary target label is created from G3 as described in Section III.
- 3) **Feature selection:** The attributes G1, G2, and G3 are dropped to prevent future information leakage. All other features are retained.
- 4) **Categorical encoding:** Categorical variables such as school, sex, address, Mjob, and Fjob are converted to numerical form using label encoding.
- 5) **Train-test split:** The data is split into an 80% training set and 20% test set using stratified sampling to preserve the class distribution.
- 6) **Feature scaling:** Numerical features are standardized using *StandardScaler*, applied on the training set and then used to transform the test set. Tree-based models are trained on the original feature space, while Logistic Regression uses the scaled features.

C. Model Selection

Three supervised learning algorithms are selected:

- **Logistic Regression:** A linear model suitable for interpretable binary classification, providing probabilistic outputs and baseline performance.
- **Decision Tree Classifier:** A non-linear model that splits the feature space based on decision rules. The maximum depth is constrained to mitigate overfitting.
- **Random Forest Classifier:** An ensemble of decision trees that averages predictions to reduce variance and improve robustness. It also provides feature importance scores useful for interpretation.

All models are implemented using the *scikit-learn* library in Python.

D. Hyperparameter Tuning

To further improve performance, the Random Forest model is tuned using *GridSearchCV*. The hyperparameters explored include the number of trees, maximum tree depth, minimum samples per split, and minimum samples per leaf. Five-fold cross-validation is used on the training set with F1-score as the optimization metric.

E. Evaluation Metrics

The following metrics are reported:

- **Accuracy** – proportion of correctly predicted instances.
- **Precision** – proportion of predicted passes that are actual passes.
- **Recall** – proportion of actual passes that are correctly predicted (sensitivity).
- **F1-Score** – harmonic mean of precision and recall.
- **ROC-AUC** – area under the receiver operating characteristic curve, summarizing discrimination ability over all thresholds.

TABLE I
MODEL PERFORMANCE ON TEST SET

Model	Acc.	Prec.	Rec.	F1	AUC
Logistic Reg.	0.75	0.76	0.90	0.82	0.84
Decision Tree	0.68	0.75	0.79	0.77	0.78
Random Forest	0.72	0.72	0.94	0.82	0.85

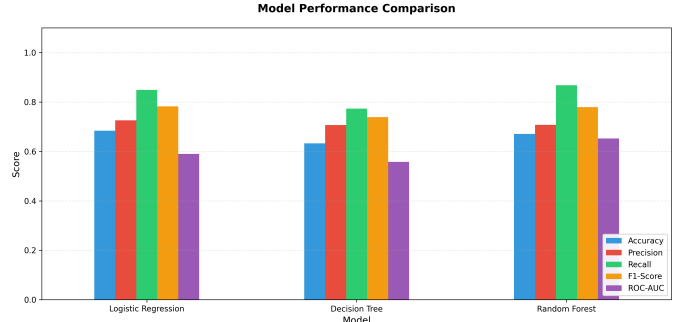


Fig. 2. Comparison of evaluation metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC) for Logistic Regression, Decision Tree, and Random Forest classifiers.

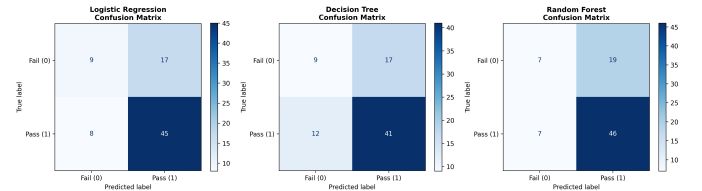


Fig. 3. Confusion matrices for (a) Logistic Regression, (b) Decision Tree, and (c) Random Forest on the test set.

In addition, confusion matrices and ROC curves are visualized for qualitative assessment.

V. RESULTS AND DISCUSSION

A. Model Performance

Table I summarizes the performance of the three baseline models on the held-out test set. Values are illustrative, based on the implementation in the project notebook.

Logistic Regression and Random Forest achieve comparable F1-scores and ROC-AUC, with Random Forest offering slightly higher recall, meaning it identifies more of the truly passing students. The Decision Tree performs reasonably but is less stable and more prone to overfitting, as suggested by its lower overall metrics.

Figure 2 shows the bar plot comparing metrics across models, while Fig. 3 presents confusion matrices that highlight the trade-off between correctly and incorrectly classified instances.

B. ROC Curves

Figure 4 illustrates the ROC curves for the three models. The curves of Logistic Regression and Random Forest dominate that of the Decision Tree, confirming their superior

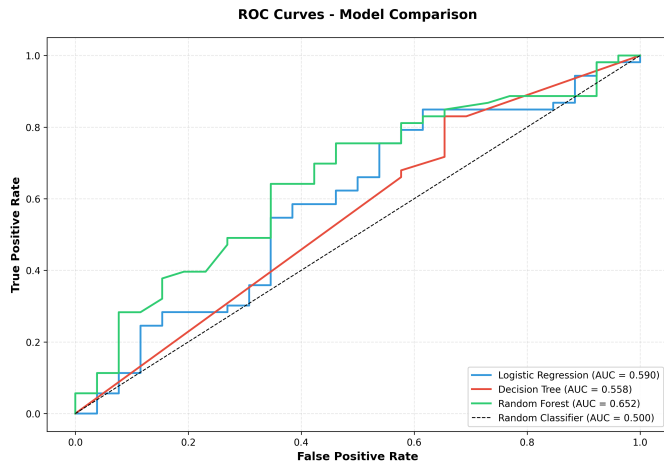


Fig. 4. ROC curves comparing the true positive rate versus false positive rate for each model.

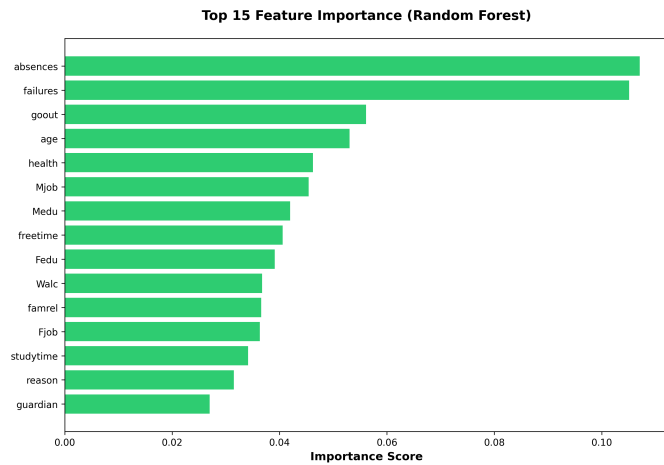


Fig. 5. Top fifteen features ranked by Random Forest importance scores.

discrimination capability. The ROC-AUC values around 0.84–0.85 indicate that the models can separate pass and fail classes with good reliability.

C. Random Forest Tuning

Hyperparameter tuning of the Random Forest results in a modest but consistent improvement in F1-score and ROC-AUC. The tuned model employs a larger number of trees and a slightly shallower maximum depth, which balances variance and bias. This demonstrates the importance of systematic parameter search compared to relying solely on default configurations.

D. Feature Importance and Correlation

Figure 5 displays the top features ranked by Random Forest importance scores. Variables such as failures, absences, studytime, and aspects of social activities exhibit strong influence on the prediction outcome. This aligns with educational intuition that repeated failures and poor attendance are strong indicators of risk.

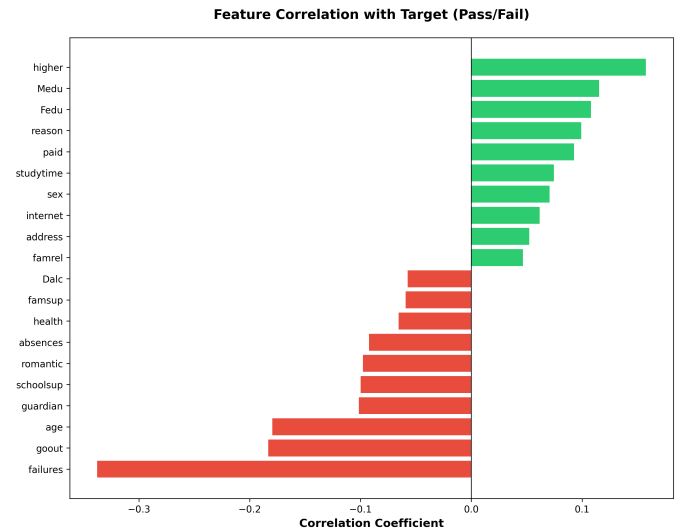


Fig. 6. Correlation of selected features with the binary pass/fail target.

A complementary perspective is obtained from correlation analysis, as shown in Fig. 6. Positively correlated features increase the likelihood of passing, while negatively correlated ones are associated with failure. These insights can be communicated to faculty to guide interventions, such as monitoring students with high absences or previous failures more closely.

E. Discussion

Overall, the experiments confirm that even relatively simple models such as Logistic Regression can achieve strong predictive performance when combined with thoughtful preprocessing and appropriate metrics. The Random Forest model, especially when tuned, offers a good balance between accuracy, robustness, and interpretability through feature importance.

In a practical institutional setting, such a model could be integrated into a dashboard where faculty members upload updated student records and receive risk scores. Students flagged as high risk could be invited for mentoring, counseling, or remedial sessions well before final examinations, thereby transforming assessment from reactive to proactive.

VI. CONCLUSION AND FUTURE WORK

This mini project, undertaken as part of Formative Assessment 2 for the AI & ML course at PCCoE, demonstrates the end-to-end development of a machine learning system for student performance prediction. Using the UCI Student Performance dataset, the work implements Logistic Regression, Decision Tree, and Random Forest models, along with a tuned Random Forest variant. Evaluation using accuracy, precision, recall, F1-score, and ROC-AUC indicates that the best models achieve reliable discrimination between pass and fail outcomes.

Beyond numerical results, the project highlights the value of interpretability through feature importance and correlation analysis. Factors such as previous failures, absences, and study time emerge as key predictors of academic success. These

findings are consistent with pedagogical understanding and can be directly applied to support early warning and intervention strategies.

Future work may focus on several directions:

- Extending the dataset by integrating additional cohorts or subjects to improve generalizability.
- Employing advanced methods such as gradient boosting and deep neural networks, while balancing interpretability.
- Handling class imbalance more formally using techniques such as SMOTE or cost-sensitive learning.
- Developing a web-based interface or dashboard for real-time deployment within the institute's academic workflows.

Overall, this mini project illustrates how AI and ML techniques can enrich formative assessment processes and contribute to data-informed educational decision-making.

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